INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM

TECH AFFAIRS
IIITDM

CAD MODELLING & INDUSTRY-ALIGNED TOOLS

Fusion 360 for:

- **CAD- Computer Aided Design,**
- **CAE- Computer Aided Engineering,**
- **CAM- Computer Aided Manufacturing.**

SAE CLUB
IIITDM KANCHEEPURAM

Society of Automotive Engineers-
SAE Collegiate Club

Session by
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Motivation?

Why is there an emphasis on CAD and related industry tools?

- Every product starts with design. From a water bottle to a cruise ship, Computer Aided Design is used to visualise, test, and refine the idea before it's made.
- In India and globally, industries are transitioning to digital-first, simulation-driven design.
- Companies are looking for engineers and designers who not only imagine great solutions, but also model and test them virtually.
- Learning this provides a competitive edge in placements, projects, and research, and opens doors in prominent fields such as automotive, aerospace, robotics, and more.

So, digital design and modelling have become a language of innovation across the world, in the current era of Industry 4.0.

Prerequisites:

- Curiosity to build and explore
- Willingness to iterate and fail and get back to work
- PC, Mouse, Fusion 360 Free Version
- Understanding of 3D space, basic geometry, engg drawing- taught in Engineering Graphics Course.

Industry Design Workflow:

The step-by-step process of Ideation, Development, Testing and Preparation for manufacturing of any product, involving specific tools and skillsets (mainly, CAD, CAE, CAM) to ensure the product meets functional, aesthetic and production goals. Standard industry design workflow is as follows:

- Concept Design
- **CAD- 3D Modelling**
- **CAE- Simulation and Analysis**
- **DFM- Design For Manufacturing**
- **CAM- Programming**
- Prototyping/ Initial Manufacturing
- Physical Testing and Validation
- Design Refinement and Feedback Loop
- Final Production and Documentation
- Quality Assurance and Inspection

What is CAD? *Ans: designing an object.*

- **What is does:** Create and visualize 2D/3D models.
- **What we use it for:** Modelling geometries, create assembly from parts, make standardized drawings.
- **What is its output:** A 3D Model.
- **Who are the users:** Designers, Product Engineers.
- **When it's used:** In the early design stage of product development.
- **What is the goal:** Define the physical properties of the product.

What is CAE? *Ans: checking if the object will survive and perform under real-life conditions.*

- **What is does:** Simulate real-time performance, test failure scenarios and optimize product designs.
- **What we use it for:** FEA (Finite Element Analysis), CFD (Computational Fluid Dynamics), thermal vibration, crash optimization.
- **What is its output:** Simulation results.
- **Who are the users:** Analysts, Structural Engineers, CFD Engineers.
- **When it’s used:** In the mid stage to validate or iterate the design model.
- **What is the goal:** Ensure the design works safely and efficiently in real-time.

What is DFM? *Ans: to make the object easily and affordably (Design For Manufacturing)*

- **What is does:** Editing/ Designing with manufacturing feasibility in mind.
- **What we use it for:** Simplifying the design model for easier production, primarily by using GD&T (Geometric Dimensioning & Tolerancing) method.
- **What is its output:** Changes in the design for optimization.
- **Who are the users:** Design Engineers, Manufacturing Engineers.
- **When it’s used:** Between CAD and CAM Stages.
- **What is the goal:** Ensure the product can be made cost-effectively.

The required tools for DFM are built in CAD Software

What is CAM? *Ans: Building the object using machines.*

- **What is does:** Convert design models into physical products
- **What we use it for:** toolpath generation, generate codes for CNC (Computer Numerical Control), 3D Printing.
- **What is its output:** Machine readable instructions.
- **Who are the users:** Manufacturing Engineers, Machinists.
- **When it’s used:** Final stage; before physical production.
- **What is the goal:** Ensure the design model can be made efficiently.

Widely used Software/ Tools:

	Global Companies *(BMW, Lockheed Martin, Samsung, Siemens)*	Indian Companies *(TATA, ISRO, TVS Motors, Tech Mahindra)*	Indian Startups *(Agnikul Cosmos, Ultraviolette, Zypp Electric)*	Students/ Non-Commercial (Free)
CAD & DFM	CATIA, Siemens NX, PTC Creo, SolidWorks	CATIA, NX, Creo, SolidWorks	Fusion 360, onshape	Fusion 360, onshape, FreeCAD, SolidWorks Student
CAE	ANSYS, Abaqus, Siemens Simcenter	ANSYS, Altair, COMSOL	ANSYS, Fusion 360 Simulation, SimScale	Fusion 360 Simulation, ANSYS Student, SimScale
CAM	MasterCAM, Siemens NX CAM, Autodesk PowerMill	NX CAM, Fusion CAM	Fusion 360 CAM	Fusion 360 CAM, onshape CAM, FreeCAD CAM

Other important things to know:

- **Geometric Dimensioning & Tolerancing (GD&T):**
  - Symbolic language used in engineering drawings, clearly describing the size, form, orientation and location of a part. These standardized symbols remove ambiguity.
  - Examples: Diameter- ∅, Perpendicular- ⊥, Parallel- ∥.
  - Can be seen in CAD (& DFM).
- **Model Based Definition (MBD):**
  - It is a design approach where all the manufacturing and inspection information is directly embedded into the 3D CAD model without needing separate 2D drawings.
  - Consists of dimensions, GD&T, material specifications, notes and manufacturing instructions and this data is attached directly to surfaces or features of the 3D model.

Other important things to know:

- **Product Lifecycle Management (PLM):**
  - It is a system/strategy used by companies to manage everything about a product from its initial concept to final disposal: connecting design data, revisions, processes, people, and tools
  - It acts like the central nervous system and manages the CAD->CAE->(DFM)->CAM workflow, behind the scenes
  - Why it is important: prevents data loss, confusion, manage thousands of parts, speeds up time to market
  - Software: Siemens Teamcenter, Dassault ENOVIA, Autodesk Fusion Manage, PTC Windchill

Quick Quiz?

1. In CAE, what does a "factor of safety" greater than 1 indicate?
A. The material will definitely fail
B. The part is over-designed and unsafe
C. The part is expected to survive the applied load
D. The simulation is invalid

2. In CAD, what is the role of a design tree (or feature tree)?
A. It organizes materials and surface finishes
B. It defines the tolerance stack for manufacturing
C. It stores sequential operations and allows rollback or editing
D. It tracks toolpaths for CAM operations

3. Which of the following best describes the core advantage of using constraints in 2D CAD sketching?
A. It locks the sketch from being edited further
B. It prevents all motion of sketch elements
C. It establishes relationships that control geometry behavior
D. It automatically converts 2D sketches into 3D models

Mini-Project

Bike Pedal Bracket, Drone Arm, etc.

Ideas:

- CAD
  - 2D profile sketch, define parameters, dimensions, constraints
  - Use modelling features to bring it to 3D: extrude, fillet, hole wizard
  - Create assembly, along with other interacting parts
- CAE
  - Assign material, apply fixtures, loads
  - Run Static Structural FEA
  - Check stress, deformation, FoS
- DFM
  - Add fillets, draft angles, modify hole sizes
  - Apply GD&T symbols on drawing, define tolerances
- CAM
  - Set-up: stock size, tools selection, origin; generate operations: facing, pocket milling
  - Simulate toolpath, generate G-Code

Roadmap to Learning

- Beginner (Fusion 360)
  - Learn from tutorials on CAD websites, YouTube, NPTEL, Coursera. Explore the software
  - Create a 3D Model using Sketching, Extrusion, Fillet and other Basic Modelling Features in Part files
- 3D Parametric Modelling
  - Manipulate the 3D Model using Parameters, Constraints and Relations
  - Make Assembly from Part files
- Intermediate
  - Advanced Features: Surface Modelling, Sheet Metal, Weldments, Frames
- CAE- Simulation (Fusion 360 Simulation)
  - FEA: Applying Loads, Supports; Evaluating Deformation, Stress; CFD: Airflow, Cooling, Pressure Simulations
  - Improve Meshing, Boundary Conditions
- CAM- Manufacturing (Fusion 360 CAM)
  - Learn about Toolpaths, learn to generate G-Code, learn the working of CNC machines
  - Simulate the milling of the Part

Roadmap to Learning

- Practice and Projects
  - Replicate real-world parts from mechanical devices
  - Take part in student competitions (e.g., SAE, Robotics, Hackathons)
- Standards and Documentation
  - Learn GD&T using ASME Y14.5 or ISO 5459
  - Explore BOMs (Bill of Materials) and revision control
- Certifications and Industry tools
  - Get certified: CSWA/CSWP (SolidWorks), Autodesk Certified User, Siemens NX Learning Path
  - Learn PLM basics (Teamcenter, Windchill overview)
- Staying Updated
  - Read industry blogs, join CAD forums, Discord, reddit, LinkedIn
  - Stay aware of Industry 4.0, 5.0 trends
- Specialize
  - Choose an industry: Automotive, Aerospace, etc.
  - Master a field: Surface Modelling, Simulation or CAM Manufacturing, etc.

Career Prospects

- **Certifications:**
  - CSWA, CSWP Exam Certifications- SolidWorks
  - Autodesk Certified User
  - Siemens NX Learning Certificate
  - ANSYS, Altair Course Completion Certificates
- **Higher Studies:**
  - M.Des, M.Tech in Product Design, Engineering Design
  - Research in CAD/CAE/CFD in IITs, NITs
  - Interdisciplinary Fields: Biomedical Engg, Sustainable Product Development
- **Industries and Job Roles:**
  - Automotive (TATA, Mahindra), Aerospace (ISRO), Consumer Products (Philips, Godrej), Startups (Ather, Agnikul)
  - Product Designer, CAE Analyst (FEA/CFD Specialist), DFM Consultant, CAM Programmer

Question and Answer time…
Ask your doubts

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